Python exam - Part 1 [Theory] (Version : 0)

TEST

Correct Answer

Answered in 2.42 Minutes

Question 1/18

You are contributing to an open-source Python project with a strict adherence to coding standards. According to PEP 8, how should you format statements in your code:

- [x] Use separate lines for each statement to enhance readability.
- [ ] Place multiple statements on a single line whenever possible.
- [ ] Place all variable assignment statements on the same line.
- [ ] Separate multiple statements on a single line with the & character.

Explanation:

1. **Use separate lines for each statement to enhance readability** is correct: Aligns with PEP 8's emphasis on readability, making code more maintainable and understandable.
2. Place multiple statements on a single line whenever possible is incorrect: Contradicts PEP 8 guidelines which recommend readability over compactness.
3. Separate multiple statements on a single line with the & character is incorrect: The & character is used for bitwise operations, not statement separation, showing a misunderstanding of both syntax and PEP 8 guidelines.
4. Place all variable assignment statements on

the same line is incorrect: Goes against PEP 8 recommendations for maintaining readability by spreading out assignments for clarity.

Question 2/18

A data scientist uses a lambda function to quickly transform a list containing E. coli levels on various days into a list of categories ('High', 'Medium', 'Low'). What is a key characteristic of lambda functions that makes them suitable for this task?

- [x] They allow for concise one-line function definitions.
- [ ] They are designed for complex multi-line operations.
- [ ] They are more efficient because they are not interpretable.
- [ ] They can iterate through lists without being defined in the traditional manner.

Explanation:

1. **They allow for concise one-line function definitions is** correct: Lambda functions are ideal for simple, one-time-use functions, like transforming list elements, due to their concise syntax.
2. They can iterate through lists without being defined in the traditional manner is incorrect: While lambda functions simplify function definition, iteration over lists typically requires an external function like map().
3. They are more efficient because they are not interpretable is incorrect: Lambda functions are interpreted, not compiled, and their efficiency comes from their succinctness, not execution speed.
4. They are designed for complex multi-line operations is incorrect: Lambda functions are designed for simplicity and brevity, not suited for complex operations that require multiple lines.

Question 3/18

Which of the following would separate a string, input_string, on the first 2 occurrences of the letter “e”?

- [ ] split(input_string == 'e', 2)
- [x] input_string.split('e', maxsplit=2)
- [ ] split(input_string = 'e', 2)
- [ ] 'e'.split(input_string, 2)

Explanation:

1. **input_string.split('e', maxsplit=2)**is correct: Utilises the split() method appropriately to divide the string at the first 2 occurrences of 'e'.
2. 'e'.split(input_string, 2)is incorrect: Syntax misapplied; split() is a string method, not called with the string as an argument.
3. split(input_string == 'e', 2)is incorrect: Incorrect syntax and logical structure, showing a misunderstanding of function calling and parameter passing.
4. split(input_string = 'e', 2)is incorrect: Syntax error with assignment; split() does not use assignment in its parameters.

Question 4/18

While working on a text analysis tool, you need to modify a string based on user input dynamically. Given Python's string immutability, what would be the most efficient way to approach this?

- [ ] Use a mutable string type available in Python.
- [ ] Directly modify the string in place.
- [ ] Convert the string to a list, modify it, then convert it back.

- [x] Create a new string with the required modifications.

Explanation:

1. **Create a new string with the required modifications** is correct: Reflects the proper approach by creating a new modified string, adhering to Python's string immutability.
2. Directly modify the string in place is incorrect: Python strings are immutable, making in-place modification impossible.
3. Convert the string to a list, modify it, then convert it back is incorrect: This method does work; however, it is not the most efficient way to do it, which is what the question asked.
4. Use a mutable string type available in Python is incorrect: Misunderstands Python data types; there are no mutable string types in Python.

Question 5/18

What is the output for:

S = [['him', 'sell'], [90, 28, 43]]

S[0][1][1]

- [ ] '28'
- [ ] '8'
- [x] 'e'
- [ ] 'i'

Explanation:

1. **'e'** is correct: Accesses the second character of the string 'sell' in the first sublist.
2. 'i' is incorrect: Miss identified the indexes.
3. '28' is incorrect: Miss identified the indexes.
4. '8' is incorrect: Miss identified the indexes.

Question 6/18

What happens if the base condition isn't defined in recursive programs?

- [x] Program goes into an infinite loop
- [ ] An exception is always thrown
- [ ] Program runs once
- [ ] Program runs n number of times where n is the argument given to the recursive function

Explanation:

1. **Program goes into an infinite loop** is Correct: Accurately describes the critical role of a base condition in preventing infinite recursion.
2. Program runs once is Incorrect: Misunderstands the functioning of recursion without a base condition.
3. Program runs n number of times where n is the argument given to the recursive function is Incorrect: Incorrect assumption about the recursion's execution without a termination condition.
4. An exception is always thrown is Incorrect: Misrepresents the outcome of missing a base condition in recursion, not necessarily leading to an exception.

Question 7/18

When writing a unit test in Python for a function, calculate_sum, that returns the sum of two integers, which of the following is the correct way to assert that the function returns 10 when passed the values 7 and 3?

- [ ] assertEquals(calculate_sum[7, 3], 10)
- [x] assert calculate_sum(7, 3) == 10

- [ ] assertEqual calculate_sum(7, 3) == 10

- [ ] calculate_sum(7, 3).assertEqual(10)

Explanation:

1. **assert calculate_sum(7, 3) == 10** is Correct: This statement correctly uses the built-in assert to directly compare the output of calculate_sum with the expected value, adhering to Python's syntax and testing conventions.
2. calculate_sum(7, 3).assertEqual(10) is Incorrect: This approach mistakenly tries to invoke assertEqual on the result of calculate_sum(7, 3), which is a syntax error since assertEqual is a method of unittest.TestCase and cannot be used in this manner.
3. assertEqual calculate_sum(7, 3) == 10 is Incorrect: This option is syntactically incorrect because it omits the necessary parentheses for calling assertEqual and incorrectly suggests it can be used without referencing a unittest.TestCase instance. It also improperly combines it with an == comparison, which is not how assertEqual is used.
4. assertEquals(calculate_sum[7, 3], 10) is Incorrect: This option is incorrect for two reasons: it misuses assertEquals, a method that should be called on a unittest.TestCase instance (and is also deprecated in favour of assertEqual), and it incorrectly attempts to call the calculate_sum function using square brackets instead of parentheses.

Question 8/18

What is instantiation in terms of OOP terminology?

- [ ] Copying an instance of class

- [ ] Deleting an instance of class

- [x] Creating an instance of class

- [ ] Modifying an instance of class

Explanation:

a. Creating an instance of class is Correct: Accurately defines instantiation as creating an object from a class.
b. Modifying an instance of class is Incorrect: Modifying an instance occurs post-instantiation and is unrelated to the instantiation process.
c. Copying an instance of class is Incorrect: Copying an instance is different from instantiation, focusing on duplicating existing objects.
d. Deleting an instance of class is Incorrect: Deletion pertains to removing an instance, not creating one.

Question 9/18

An environmental scientist aims to combat climate change by analysing a dataset of yearly carbon dioxide emissions per country using the pandas library. The objective is to sort the data to highlight the top emitting countries. Which pandas method correctly sorts the DataFrame emissions by the column CO2_Emissions in descending order?

- [ ] emissions.pivot_table(index='CO2_Emissions', aggfunc='sum').sort_values('CO2_Emissions', ascending=False)
- [x] emissions.sort_values(by='CO2_Emissions', ascending=False)
- [ ] emissions.groupby('CO2_Emissions').sort_values('CO2_Emissions', ascending=False)
- [ ] emissions.sort_values(by='CO2_Emissions', descending=False

Explanation:

a. emissions.sort_values(by='CO2_Emissions', ascending=False) is Correct: sort_values method correctly sorts the DataFrame by the column CO2_Emissions in descending order.
b. emissions.groupby('CO2_Emissions').sort_values('CO2_Emissions', ascending=False) is Incorrect: sort_values is not a

method of a grouped DataFrame, leading to an AttributeError.
c. emissions.pivot_table(index='CO2_Emissions', aggfunc='sum').sort_values('CO2_Emissions', ascending=False)is Incorrect: pivot_table and sort_values don't sort the original DataFrame by CO2_Emissions.
d. emissions.sort_values(by='CO2_Emissions', descending=False is Incorrect: sorts the DataFrame by the column CO2_Emissions in ascending order.

Question 10/18

___ works by taking elements from the unsorted list and inserting them at the right place in a new sorted list.

- [x] Insertion sort
- [ ] Selection sort
- [ ] Quick sort
- [ ] Bubble sort

Explanation:

a. Insertion sort is Correct: Accurately describes Insertion sort as gradually building up a sorted array by inserting elements in the correct position.
b. Selection sort is Incorrect: Misidentifies Selection sort's mechanism, which involves finding the minimum element and placing it at the beginning.
c. Quick sort is Incorrect: Incorrectly attributes the mechanism of Quick sort to the gradual insertion process.
d. Bubble sort is Incorrect: Misunderstands Bubble sort, which repeatedly swaps adjacent elements to sort a list, not inserting them in a new list.

Question 11/18

The time complexity of a quick sort algorithm, which makes use of median, found by an O(n) algorithm, as pivot element is:

- [ ] O(logn)
- [ ] O(n)
- [ ] O(n^2)
- [x] O(nlogn)

Explanation:

1. **O(nlogn)** is Correct: Correctly identifies the time complexity of Quick sort under optimal conditions using median as pivot.
2. O(n^2) is Incorrect: Represents the worst-case scenario, not the general complexity with optimal pivot selection.
3. O(logn) is Incorrect: There is no version of Quick Sort or method of pivot selection that brings its time complexity down to O(logn) for sorting an array.
4. O(n) is Incorrect: Misrepresents Quick sort's complexity, suggesting a linear time which is not achievable for sorting.

Question 12/18

An environmental scientist is developing a model to simulate the regrowth of deforested areas over time. The model needs to predict the area's recovery through a function that estimates growth each year based on the previous year's growth, creating a chain of function calls until a target year is reached. What programming concept is being applied here?

- [ ] Iteration, using loops to repeatedly execute a block of code.
- [x] Recursion, where a function calls itself with a modified parameter until a base case is reached.
- [ ] Polymorphism, where a function operates in multiple ways depending on the input.
- [ ] Inheritance, where a class inherits

properties from a parent class.

Explanation:

1. **Recursion, where a function calls itself with a modified parameter until a base case is reached** is Correct: Correctly identifies recursion as the concept applied for predicting regrowth through self-referential function calls.
2. Iteration, using loops to repeatedly execute a block of code is Incorrect: Confuses the scenario with iteration; while related to recursion, it doesn't involve self-calling functions.
3. Inheritance, where a class inherits properties from a parent class is Incorrect: Misapplies inheritance, an OOP concept, to a scenario involving function calls and predictions.
4. Polymorphism, where a function operates in multiple ways depending on the input is Incorrect: Incorrectly applies polymorphism, which is about a function's ability to process objects differently based on their class or data type.

Question 13/18

How would you add changes to a repository on GitHub using a terminal?

- [ ] git commit -m "Update file"; git add .; git push
- [x] git add .; git commit -m "Update file"; git push
- [ ] git add .; git push; git commit -m "Update file"
- [ ] git push; git add .; git commit -m "Update file"

Explanation:

1. **git add .; git commit -m "Update file"; git push** is Correct: Accurately outlines the sequence of Git commands for staging,

committing, and pushing changes.

2. git commit -m "Update file"; git add .; git push is Incorrect: Misplaces the order of Git commands, showing a misunderstanding of the correct workflow.
3. git push; git add .; git commit -m "Update file" is Incorrect: Starts with git push, which is incorrect as changes need to be staged and committed first.
4. git add .; git push; git commit -m "Update file" is Incorrect: Incorrectly sequences the commit and push operations; commits should precede pushes.

Question 14/18

How is the variance for a sample standard deviation calculated?

- [ ] By summing the squared differences between each observation and the population mean, then dividing by the number of sample observations minus one (n−1).
- [ ] By summing the squared differences between each observation and the sample mean, then dividing by the number of population observations plus one (N+1).
- [ ] By summing the squared differences between each observation and the sample mean, then dividing by the number of population observations (N).
- [x] By summing the squared differences between each observation and the sample mean, then dividing by the number of sample observations minus one (n−1).

Explanation:

1. Correct: By summing the squared differences between each observation and the sample mean, then dividing by the number of observations minus one (n−1).
2. By summing the squared differences between

each observation and the sample mean, then dividing by the number of population observations (N) is Incorrect: This option describes the formula for calculating the variance of a population, not a sample. When dealing with a sample, the correct denominator is n−1 instead of N. This adjustment, known as Bessel's correction, is made to produce an unbiased estimator of the population variance.

3. By summing the squared differences between each observation and the population mean, then dividing by the number of sample observations minus one (n−1) is Incorrect: This option incorrectly suggests using the population mean in the calculation of sample variance. The sample variance is calculated using the sample mean, not the population mean. The focus on the population mean misdirects from the correct procedure of using sample-specific metrics.
4. By summing the squared differences between each observation and the sample mean, then dividing by the number of population observations plus one (N+1) is Incorrect: This option introduces an incorrect formula by suggesting the denominator be N+1, which is not a recognised statistical method for calculating sample variance or standard deviation. The correct approach for sample variance involves dividing by n−1, which corrects for the bias in estimating a population parameter from a sample.

Question 15/18

Examine the following Python code intended to create a line plot to visualise the effect of temperature on maize yield. Identify any error/s in the code:

```
import seaborn as sns

import matplotlib.pyplot as plt

import pandas as pd

data = pd.DataFrame({

    'Yield_of_maize_kgs': [250, 300, 350, 400, 450],

    'Average_temperature': [18, 30, 21, 25, 27]

})

sns.scatterplot(x='Yield_of_maize_kgs', y='Average_temperature', data)
```

```
plt.title('Effect of temperature on maize yield')

plt.show()
```

- [ ] The line of code should be corrected to: sns.scatterplot(x='Yield_of_maize_kgs', y='Average_temperature', data = data)
- [x] The line of code should be corrected to: sns.lineplot(x='Average_temperature', y='Yield_of_maize_kgs', data = data)
- [ ] There is no error in the code.
- [ ] The line of code should be corrected to: sns.lineplot(x='Average_temperature', y='Yield_of_maize_kgs', data)

Explanation:

1. **sns.lineplot(x='Average_temperature', y='Yield_of_maize_kgs', data = data)** Correct: Accurately identifies the need to correct the plot type to sns.lineplot and correctly sets the axes.
2. sns.scatterplot(x='Yield_of_maize_kgs', y='Average_temperature', data = data) Incorrect: Corrects to sns.lineplot but incorrectly maintains the axes, misrepresenting the data relationship.
3. sns.lineplot(x='Average_temperature', y='Yield_of_maize_kgs', data) Incorrect: Attempts to correct the plot type but misplaces the data argument, indicating a syntax misunderstanding.
4. There is no error in the code - Incorrect: Fails to recognise the original code's mistake, incorrectly suggesting it's correct as is.

Question 16/18

Climate scientists are using sample data from various microclimates to predict broader climate patterns. When applying statistical methods to estimate the mean temperature increase due to global warming, which principle allows them to infer the distribution of the mean from these samples?

- [ ] The Monte Carlo method, a computational algorithm that relies on repeated random sampling to obtain results

- [ ] Pearson's correlation coefficient, which measures the linear correlation between two variables.

- [ ] The law of large numbers, which states that as a sample size grows, its mean gets closer to the average of the whole population.

- [x] The central limit theorem, which states that the sampling distribution of the sample mean will approximate a normal distribution, regardless of the population's distribution, given a sufficiently large sample size.

Explanation:

1. Correct: Accurately describes the central limit theorem's significance in estimating statistical distributions from sample data.
2. Incorrect: Misapplies the law of large numbers, which is relevant but does not directly address the question's focus.
3. Incorrect: Incorrectly applies Pearson's correlation measure to a question about statistical distribution estimation.
4. Incorrect: Misuses the Monte Carlo method, which is not directly related to the concept of sample mean distribution.

Question 17/18

In a project monitoring wildlife movements through protected areas, scientists use immutable data structures to ensure the integrity of the recorded GPS tracking data. Why is an attempt to directly modify this data type, such as updating a GPS coordinate in a tuple, problematic in Python?

- [x] Attempting to modify an immutable data structure like a tuple results in a TypeError because tuples cannot be changed after their creation.

- [ ] Immutable types do not support item assignment, making direct updates inefficient.

- [ ] Direct modification leads to a ValueError due to incorrect value assignment.

- [ ] The operation overwrites the entire data structure, leading to data loss.

Explanation:

1. **TypeError** is Correct: Accurately explains the error resulting from attempts to modify an immutable structure like a tuple.
2. ValueError is Incorrect: Incorrectly identifies the type of error associated with immutable structure modification.
3. Overwrites is Incorrect: Misunderstands how immutable data structures function, suggesting incorrect outcomes.
4. Do not support item assignment is Incorrect: Correct about item assignment support but irrelevantly mentions efficiency.

Question 18/18

Which statement is correct:

- [ ] The following code will produce a list of all the squares from 1 to 5: squares={x:x**2 for x in range(1,6)}

- [x] The following code will return False: False and True

- [ ] The following code will produce a list of squared numbers that fit between 0 to 11 where every second number is squared: [i**2 for i in range(1,11,2)]

- [ ] Tuples, dictionaries and lists are all immutable.

Explanation:

Correct: This statement will return False because it is using the logical operator "and" between False and True, which evaluates to False.

1. Incorrect: Dictionaries and lists are mutable, meaning their contents can be modified after creation.
2. Incorrect: Incorrectly identifies the result of the code snippet; squares={x:x**2 for x in range(1,6)} creates a dictionary, not a list.

d. Incorrect: This [i**2 for i in range(1,11,2)] will produce a list of squared numbers, but it won't contain every second number squared. Instead, it will contain the squares of every odd number from 1 to 10, as it's using a step of 2 in the range function.